

Database Schema & Entity Relationships

Table: AUTH_USER

Keys / Constraints	Field Name	Data Type	Notes / Limits	Description
PK	id	BigInt	Auto-increment	Primary key for the user model

Table: POST

Keys / Constraints	Field Name	Data Type	Notes / Limits	Description
PK	id	BigInt	Auto-increment	Primary key for the post
UK	title	VarChar	Max length: 200, Unique	Post title (enforced unique)
UK	slug	VarChar	Max length: 200, Unique	URL-friendly slug (enforced unique)
FK	author_id	BigInt	Not Null	Foreign key referencing AUTH_USER(id)
Not Null	content	Text	Required	Body content of the post
Not Null	created_on	DateTime	Auto-created	Set automatically when a post is

				created
Nullable	excerpt	Text	Optional	Optional summary/excerpt
Not Null	status	Integer	Default: 0	Post publication status (0: Draft, 1: Published)
Not Null	updated_on	DateTime	Auto-updated	Refreshed automatically whenever a post is saved

Table: COMMENT

Keys / Constraints	Field Name	Data Type	Notes / Limits	Description
PK	id	BigInt	Auto-increment	Auto-generated primary key
—	body	Text	—	Content of the comment
Default	approved	Boolean	Default: false	Approval status of the comment
Not Null	created_on	DateTime	Auto-created	Set automatically when comment is created
FK	author_id	BigInt	Not Null	Foreign key

				referencing AUTH_USER(id)
FK	post_id	BigInt	Not Null	Foreign key referencing POST(id)

Table: COMMENT_VOTE

Keys / Constraints	Field Name	Data Type	Notes / Limits	Description
PK	id	BigInt	Auto-increment	Primary key for the vote record
Constraint	value	SmallInt	1 or -1	Vote value (1 for Upvote, -1 for Downvote)
Not Null	created_on	DateTime	Auto-created	Set automatically when a vote is created
FK, UK	comment_id	BigInt	Not Null, Composite UK	Foreign key referencing COMMENT(id)
FK, UK	user_id	BigInt	Not Null, Composite UK	Foreign key referencing AUTH_USER(id)

2. Relationships

Source	Relationship	Target	Cardinality	On Delete	Details
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Entity	p	Entity		Action	
AUTH_USER	Authors	POST	1 : N (One-to-M any)	CASCADE	One AUTH_USER can author zero or many POST records. Every POST belongs to exactly one AUTH_USER via author_id. Deleting a user cascades to delete their posts.
AUTH_USER	Writes	COMMENT	1 : N (One-to-M any)	CASCADE	One AUTH_USER can write zero or many COMMENT records. Every COMMENT belongs to exactly one AUTH_USER via author_id. Deleting a user cascades to delete their comments.
POST	Has	COMMENT	1 : N (One-to-M	CASCADE	One POST can have

			any)		zero or many COMMENT records. Every COMMENT belongs to exactly one POST via post_id. Deleting a post cascades to delete its comments.
AUTH_USER	Casts	COMMENT_VOTE	1 : N (One-to-M any)	CASCADE	One AUTH_USER can cast zero or many COMMENT_VOTE records. Every vote belongs to exactly one user via user_id. Deleting a user cascades to delete their votes.
COMMENT	Receives	COMMENT_VOTE	1 : N (One-to-M any)	CASCADE	One COMMENT can receive zero or many COMMENT_VOTE

					records. Every vote belongs to exactly one comment via comment_id. Deleting a comment cascades to delete its votes.
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